

Basic Principles of Medicine 2
Module : Digestion and Metabolism
Lecture No: 16

Fatty Acid Synthesis and Regulation

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Lecture Objectives: Fatty Acid Synthesis and Regulation

SOM. MK.I.BPM2.3.DM.2.BCHM.1185	Outline the denovo synthesis of fatty acids highlighting the significance of acetyl CoA carboxylase.
SOM. MK.I.BPM2.3.DM.2.BCHM.1186	Identify the sources of NADPH for fatty acid synthesis.
SOM. MK.I.BPM2.3.DM.2.BCHM.1187	Outline the elongation and desaturation of fatty acids in humans.
SOM. MK.I.BPM2.3.DM.2.BCHM.1188	Explain the conversion of linoleic acid to arachidonic acid and α -linolenic acid to docosahexaenoic acid.
SOM. MK.I.BPM2.3.DM.2.BCHM.1189	List the major tissues of triacylglycerol synthesis and storage
SOM. MK.I.BPM2.3.DM.2.BCHM.1190	Distinguish between the glycerol-3-P pathway and the MAG pathway for triacylglycerol synthesis
SOM. MK.I.BPM2.3.DM.2.BCHM.1191	Explain the regulation of acetyl-CoA carboxylase (short-term and long-term) and regulation of Fatty Acid synthesis
SOM. MK.I.BPM2.3.DM.2.BCHM.1192	Discuss malonyl-CoA inhibition of carnitine-palmitoyl transferase I (CPT I).

Recommended Reading

- **Lippincott's Biochemistry**

9th Edition  <https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>

- **Chapter 16:** Fatty Acid, Triacylglycerol, and Ketone Body Metabolism - **Sections: Fatty Acid Structure and Fatty acid de novo synthesis**
 - Questions: 16.3 and 16.4
- **Chapter 13:** Pentose Phosphate Pathway and Nicotinamide Adenine Dinucleotide Phosphate - **Section: Uses of NADPH: Reductive biosynthesis**
- **Chapter 15:** Dietary Lipid Metabolism - **Section: Triacylglycerol and cholesteryl ester resynthesis**

- **Marks' Basic Medical Biochemistry:** [Marks' Basic Medical Biochemistry: A Clinical Approach, 6e](#)

- **Chapter 31:**

Sources of Fatty Acids

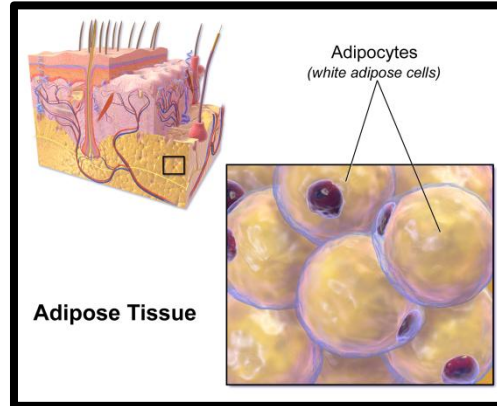
Diet



TAG

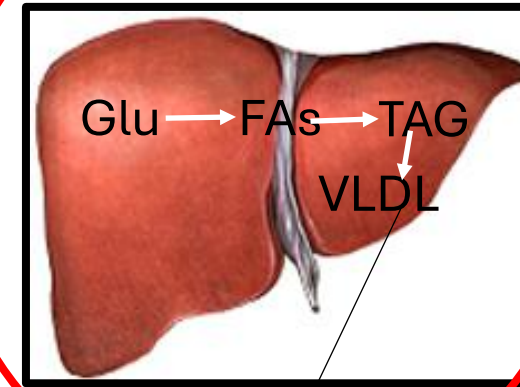
Chylomicrons

Adipose Tissue



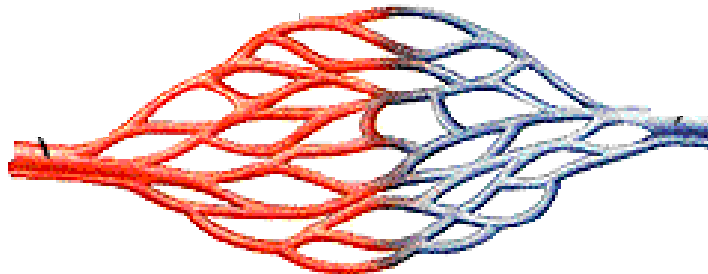
FFA

FA Biosynthesis



VLDL

Lipid transport in blood

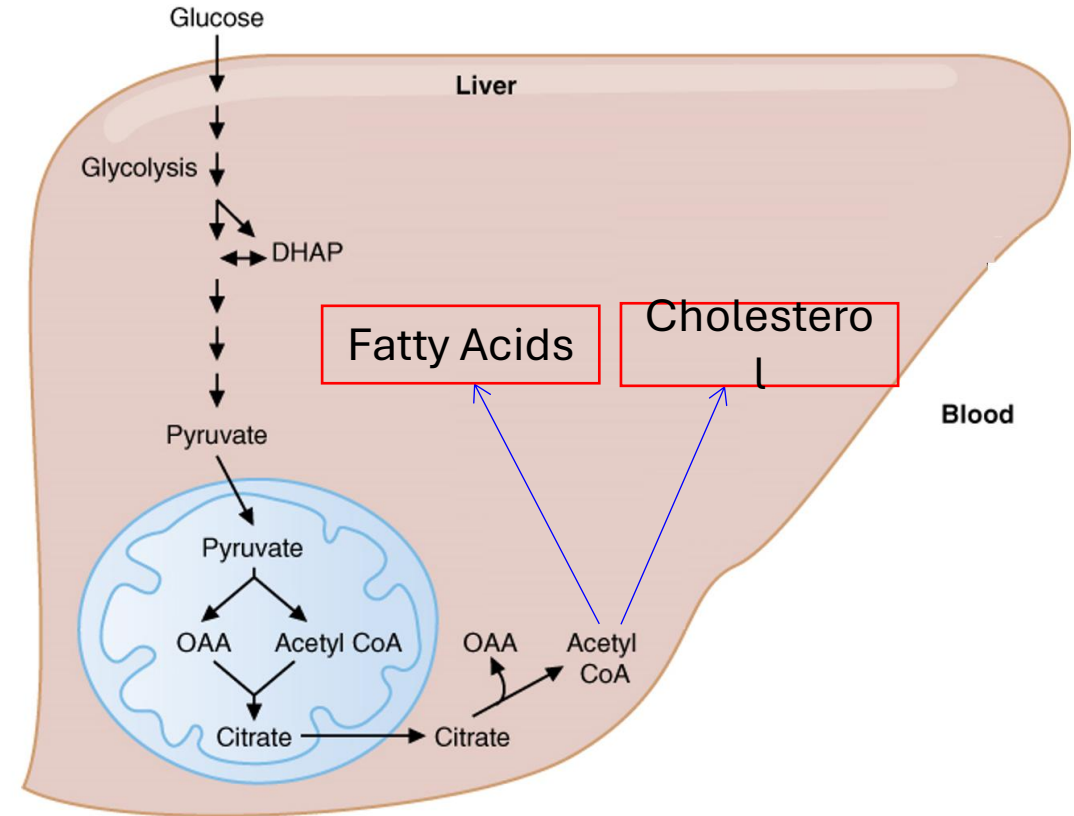


Fatty acids are obtained in liver by hepatocellular uptake from the plasma and by de novo biosynthesis.

Fatty acids play a central role in metabolism as building blocks of biological membranes, energy reserves, and precursors to second messenger molecules.

Fatty Acid Synthesis

- ✓ Fatty Acid Synthesis is Linked to Carbohydrate Metabolism.
- ✓ Sites of Biosynthesis of FAs:
 - **Major** (liver, lactating mammary gland) and
 - **Minor** (adipose tissue)
- ✓ Dysregulation of fatty acid synthesis is linked to diseases like cancer, obesity, and non-alcoholic fatty liver disease.



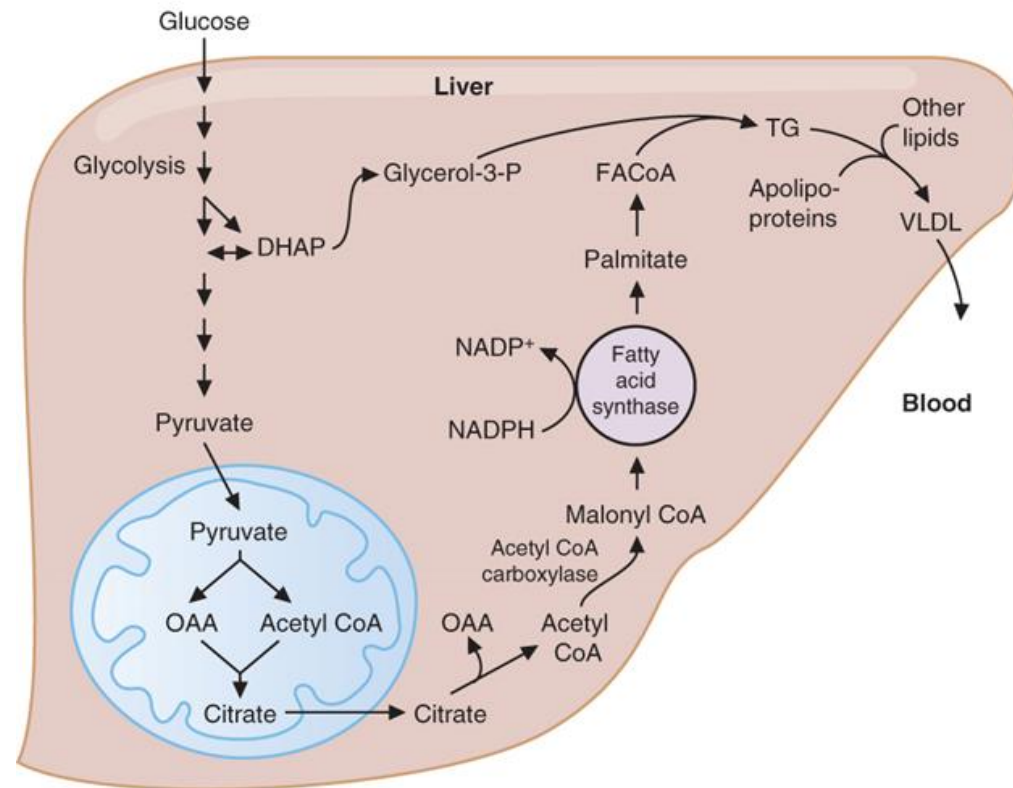
Insulin Activates Hepatic Synthesis of Fatty acids and Cholesterol

Hepatic Synthesis of Fatty Acid

✓ Site of synthesis in the cell: **cytosol**.

✓ Precursors and coenzymes

⑩ Acetyl CoA, Biotin, CoA (pantothenic acid), ATP, NADPH, CO₂

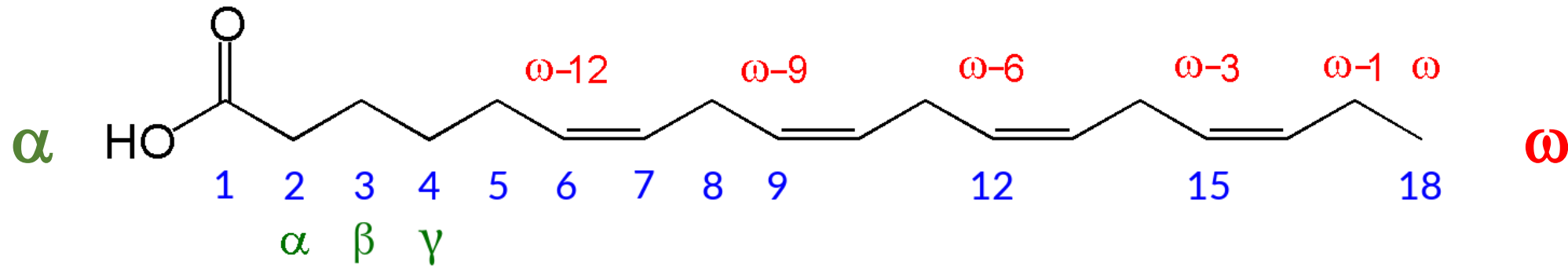


Marks' Basic Medical Biochemistry

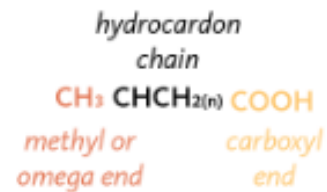
✓ Occurs in the **well-fed state (insulin)** in the **liver**.

Fatty acid nomenclature

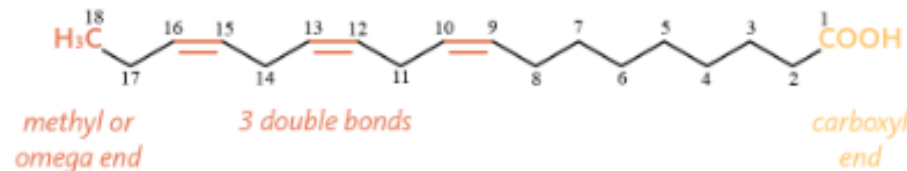
Fatty acid carbon atoms are numbered (or lettered with Greek lettering system (α , β , γ , δ) either from the carboxyl group **OR** from the carbon furthest removed from the carboxyl group (**ω numbering system**)



(a)



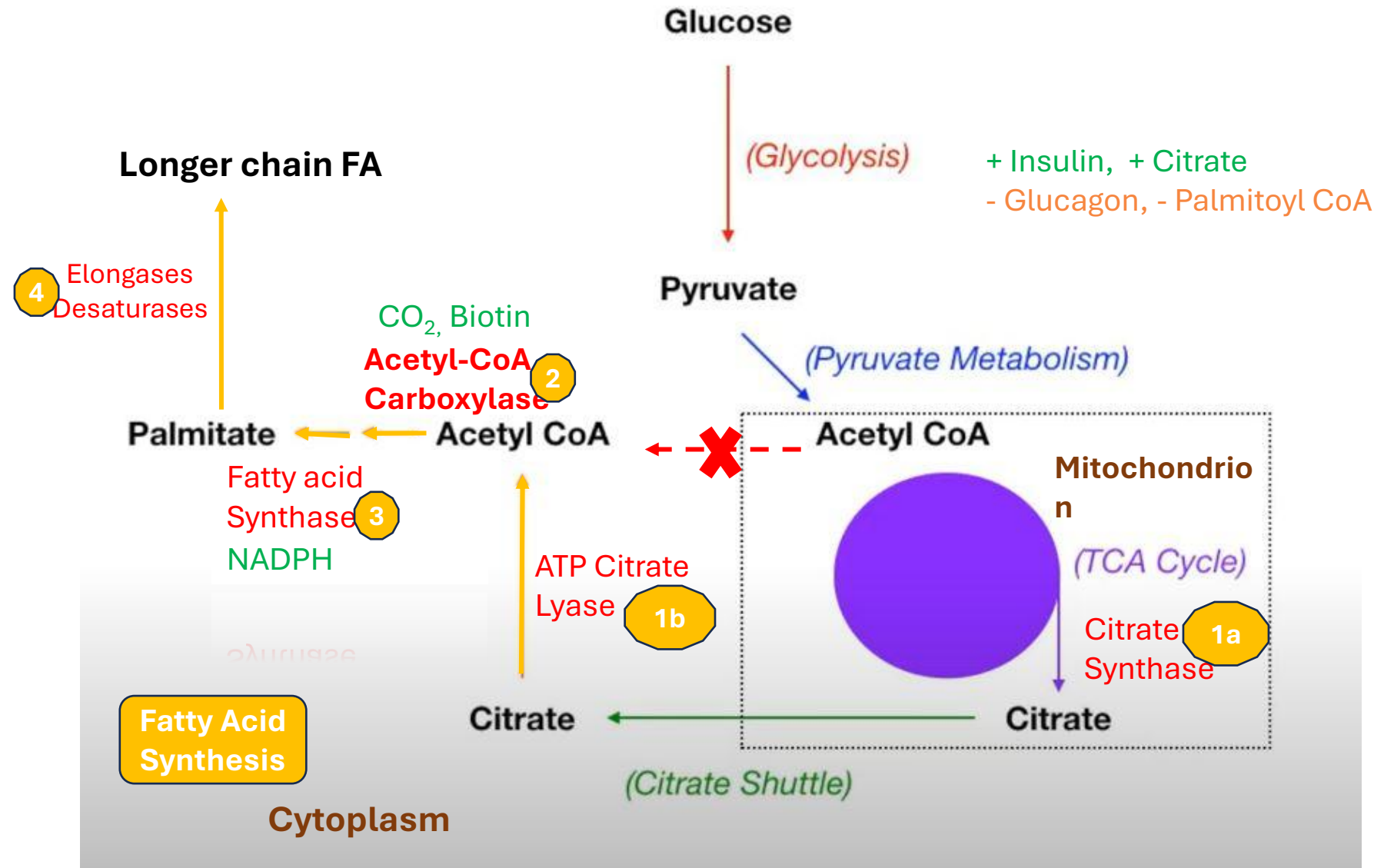
(b)



(a) The general structure of a fatty acid (22). (b) The chemical structure of α -linolenic acid (ALA), 18:3n-3. ALA has 18 carbon atoms (C) and 3 double bonds, the first of which is located 3 carbon atoms from the terminal methyl group (omega end).



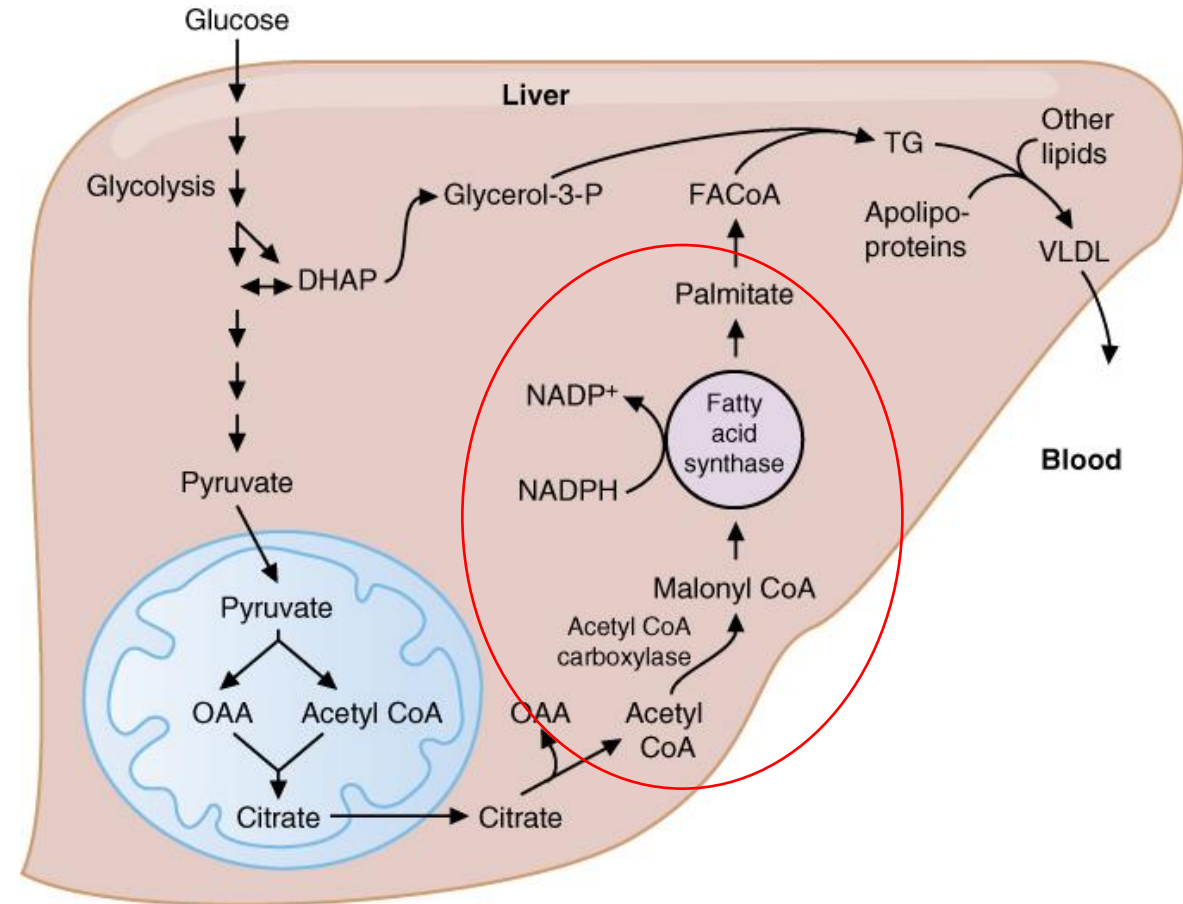
Overview of Fatty Acid Synthesis



- Excess glucose → Pyruvate → Acetyl CoA
- Acetyl CoA exits mitochondria as citrate which is then converted to acetyl-CoA by **ATP citrate lyase**.
- Acetyl-CoA → Malonyl-CoA by **acetyl-CoA carboxylase (ACC)**.
- Fatty acid synthase (FAS) condenses one acetyl-CoA and seven malonyl-CoA into palmitate which can then be modified into various fatty acids.

Mechanism of de novo Synthesis of Fatty Acids

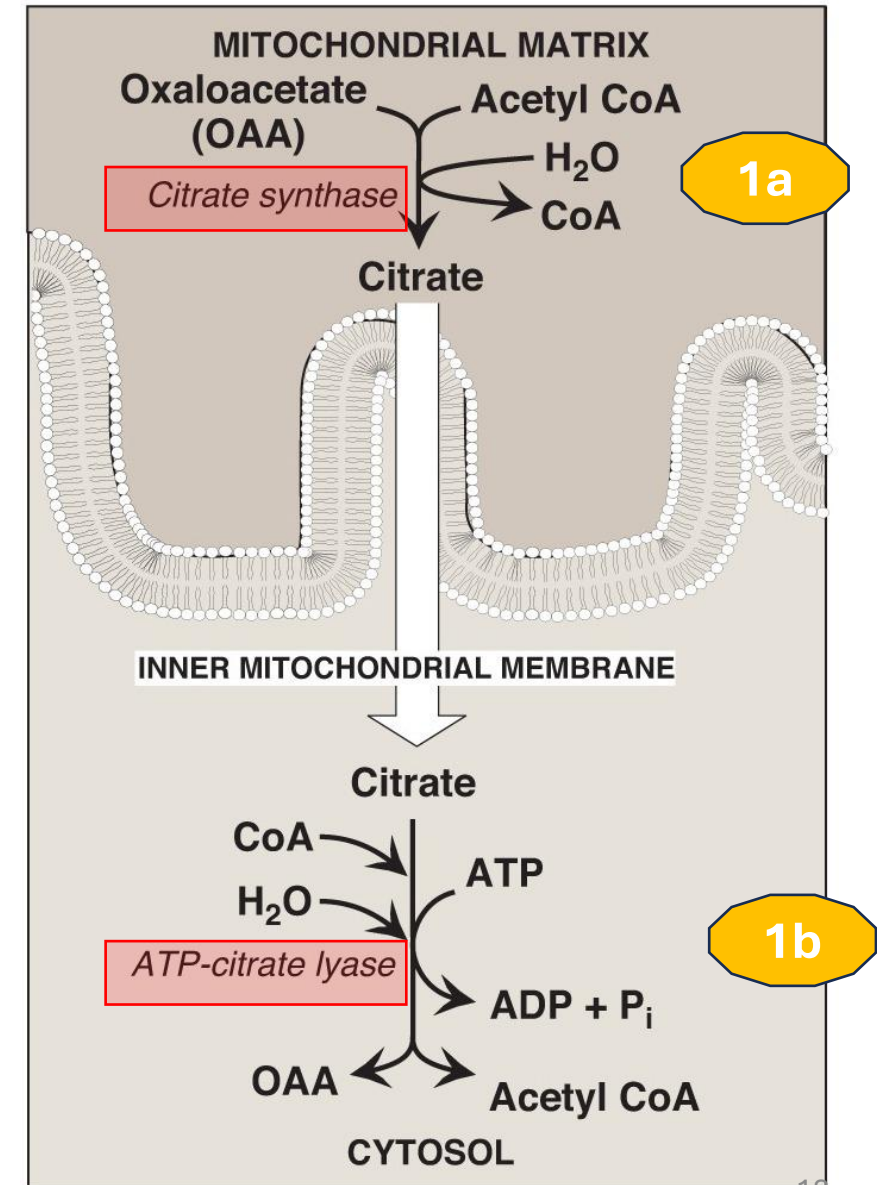
1. Production of **cytosolic acetyl CoA**.
2. Formation of **Malonyl CoA** by **acetyl CoA carboxylase**. Requires ATP, CO₂ & Biotin.
3. Synthesis of **palmitate (16C saturated FA)** by **fatty acid synthase (FAS)**. Requires NADPH
4. Modification of palmitate:
 - Elongation of palmitate by **elongases**
 - Desaturation by **desaturases**



Marks' Basic Medical Biochemistry

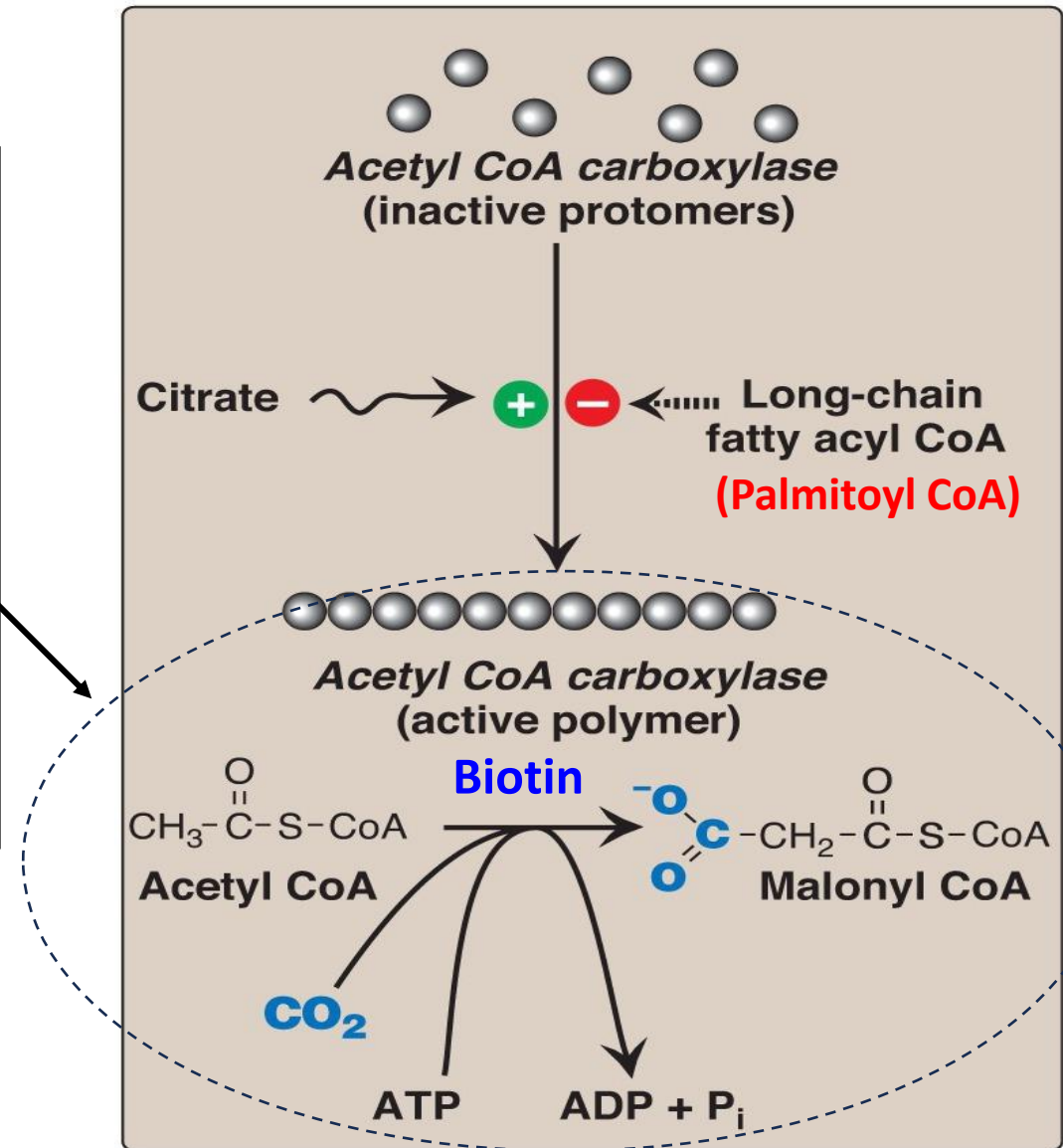
Production of Cytosolic Acetyl CoA

- Acetyl CoA formed in mitochondria cannot cross mitochondrial membrane into cytosol.
 - Acetyl CoA converted to citrate.
- Citrate leaves into the cytosol.
- ATP-Citrate lyase cleaves citrate in cytosol to form acetyl CoA and oxaloacetate.



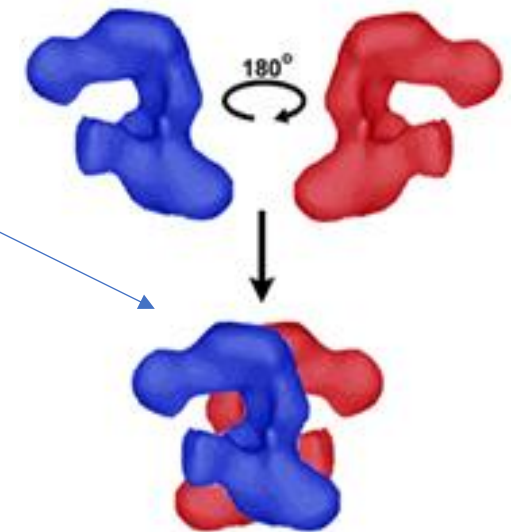
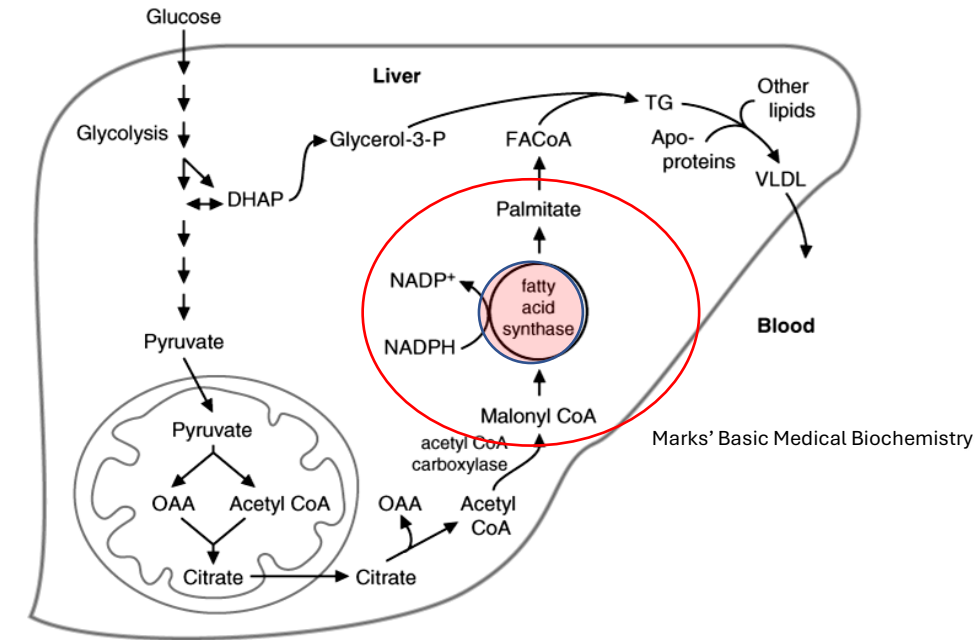
2 Acetyl CoA carboxylase (ACC): Acetyl CoA → Malonyl CoA

- Committed/ regulated step of fatty acid biosynthesis
- **Rate-limiting enzyme for fatty acid synthesis**
- **ABC enzyme: ATP; Biotin; CO₂**
 - Uses **ATP**
 - Requires **biotin** (vitamin B7)
 - Adds CO₂ to acetyl CoA → malonyl CoA.



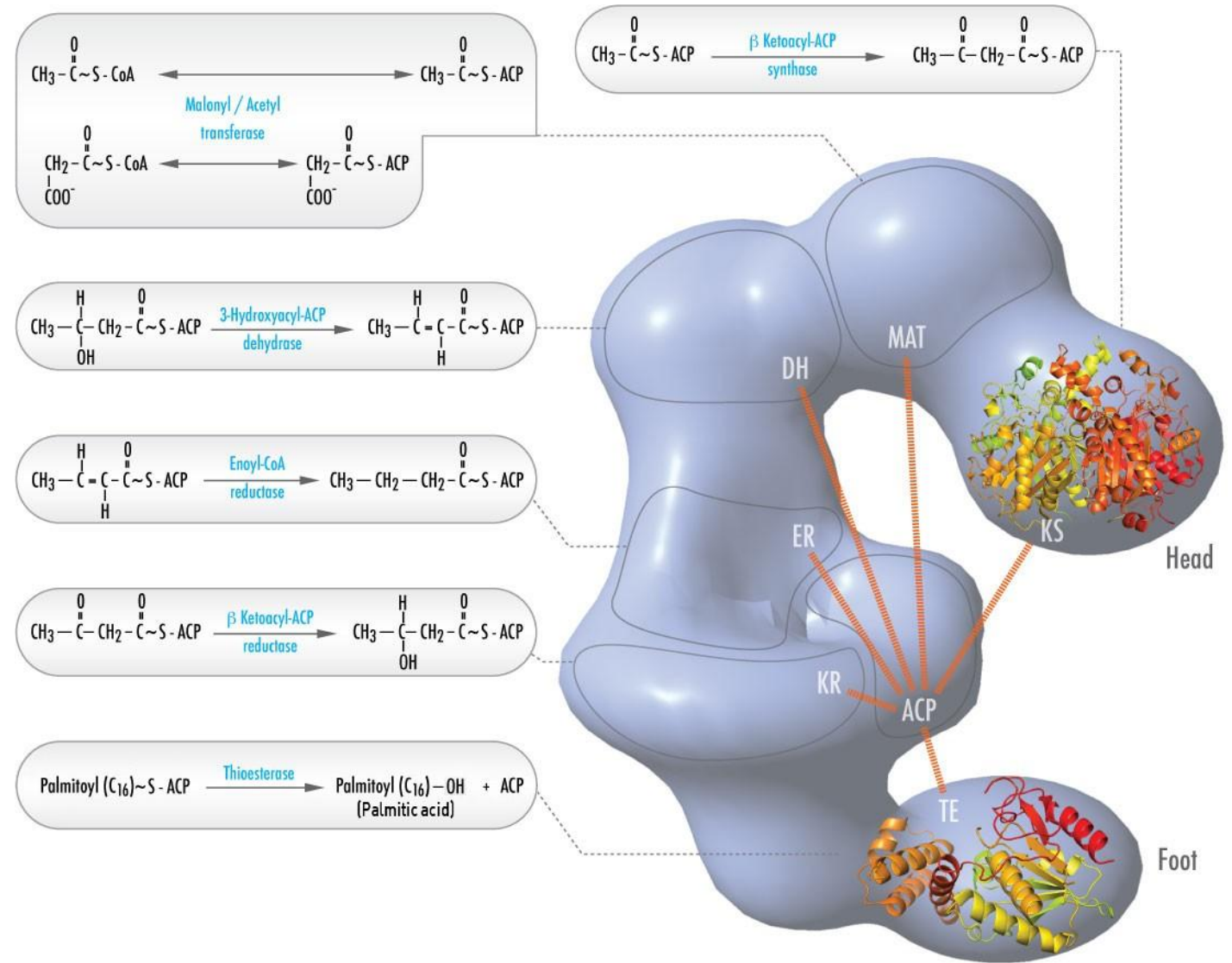
Fatty Acid Synthase complex (FAS)

- Multifunctional **FAS enzyme** catalyzes a series of reactions whereby two carbon units from **malonyl CoA** are linked together, ultimately to form **palmitoyl CoA**.
 - **Acetyl-CoA** and **7 malonyl-CoA** are condensed into 16-carbon palmitate.
- The enzyme complex exists as **dimer**
 - Each monomer has seven different enzyme activities and contains an **acyl-carrier domain (ACP)** and a temporary holding domain with a **cysteine residue**.
- **Insulin upregulates** the amount of enzyme expressed.



Fatty Acid Synthase Mechanism

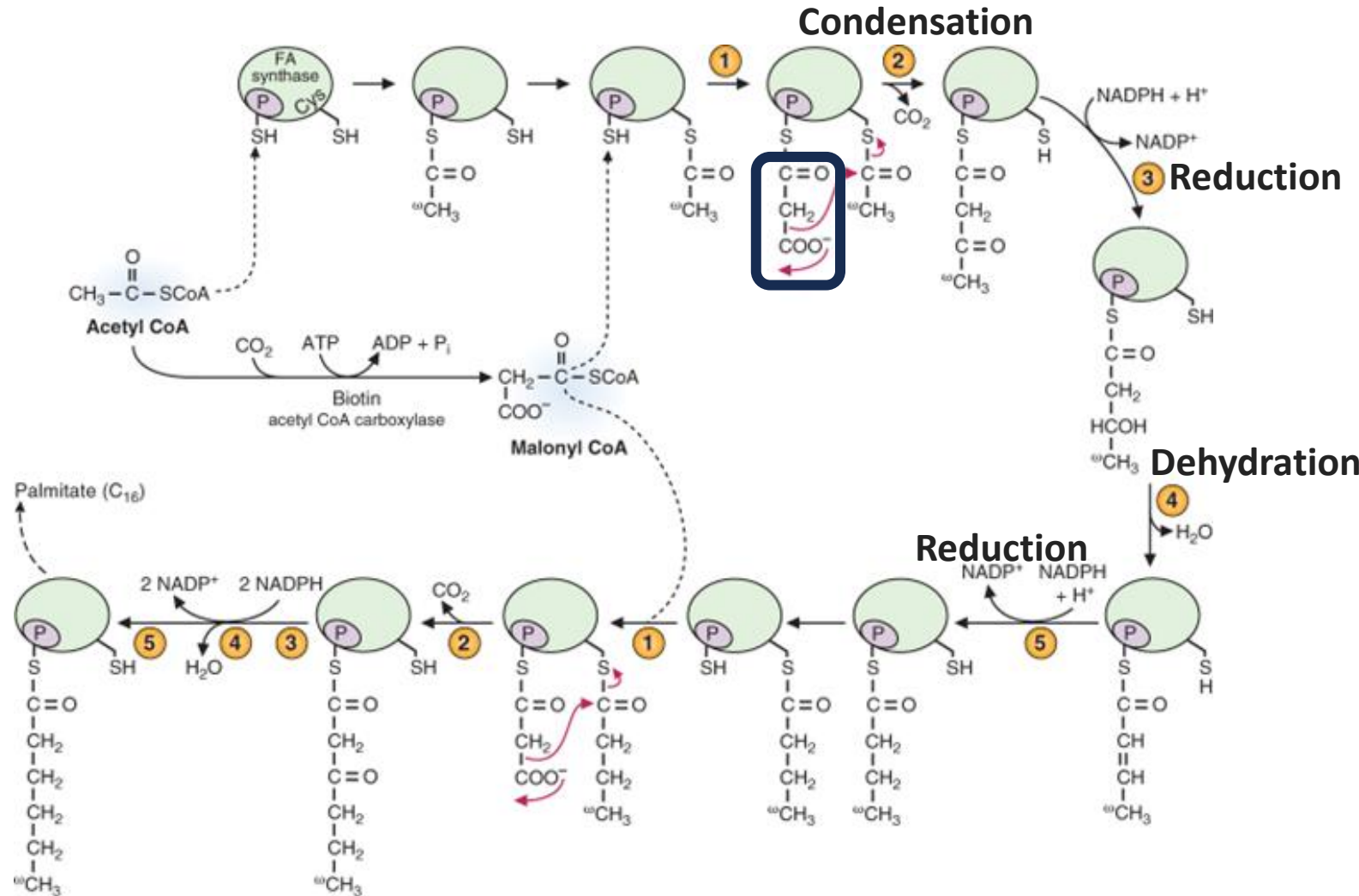
- Condensation of malonyl-CoA and acetyl-CoA
- Repeating cycle of **reduction** (Requires **NADPH**) and **dehydration** adds 2-carbons in each cycle to the fatty acid chain (elongation)
- Release palmitate (16C fatty acid) from the enzyme.



The catalytic activities of FAS complex
(Understand concept; students need not memorize the individual enzyme names)

Synthesis of Palmitate on the Fatty Acid Synthase (FAS) Complex

- Acetyl coenzyme A (acetyl CoA) is attached to FAS.
- Malonyl coenzyme A (Malonyl CoA) provides the two-carbon units that are added to growing fatty acyl chain
- **Following steps are repeated until palmitate (16 C) is produced:**
 - (1) Addition of malonyl group to the phosphopantotheinyl residue
 - (2) Condensation of malonyl and fatty acyl groups
 - (3) **Reduction** of β -ketoacyl group
 - (4) Dehydration
 - (5) **Reduction** of the double bond
- Steps 3 and 5: Use **NADPH+H⁺**



Adapted from: [Marks' Basic Medical Biochemistry: A Clinical Approach, 6e](#)

Key: ADP, adenosine diphosphate; ATP, adenosine triphosphate; Cys-SH, a cysteinyl residue on a different subunit of the fatty acid synthase; NADP, nicotinamide adenine dinucleotide phosphate; P, a phosphopantetheinyl group attached to the fatty acid synthase complex; P_i, inorganic phosphate.

Sources of NADPH for Fatty Acid Synthesis

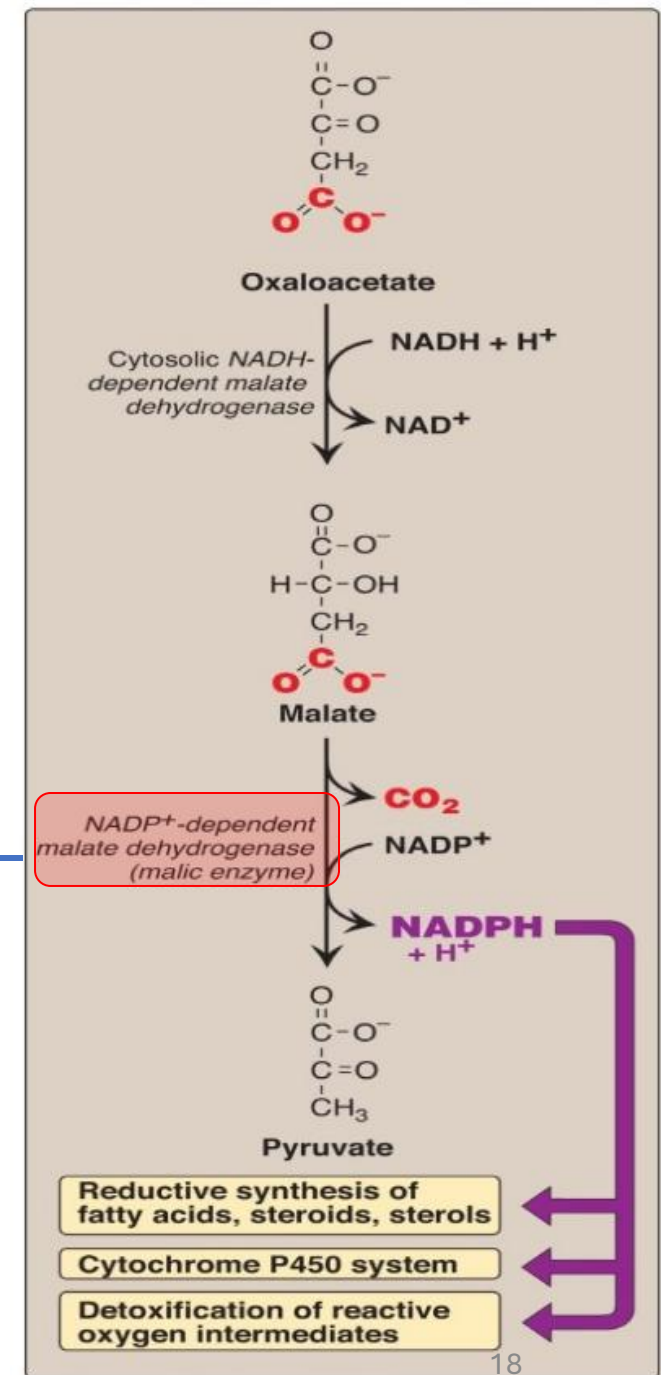
NADPH required to provide reducing equivalents for FA biosynthesis is produced by:

1. Pentose phosphate pathway (Major source)

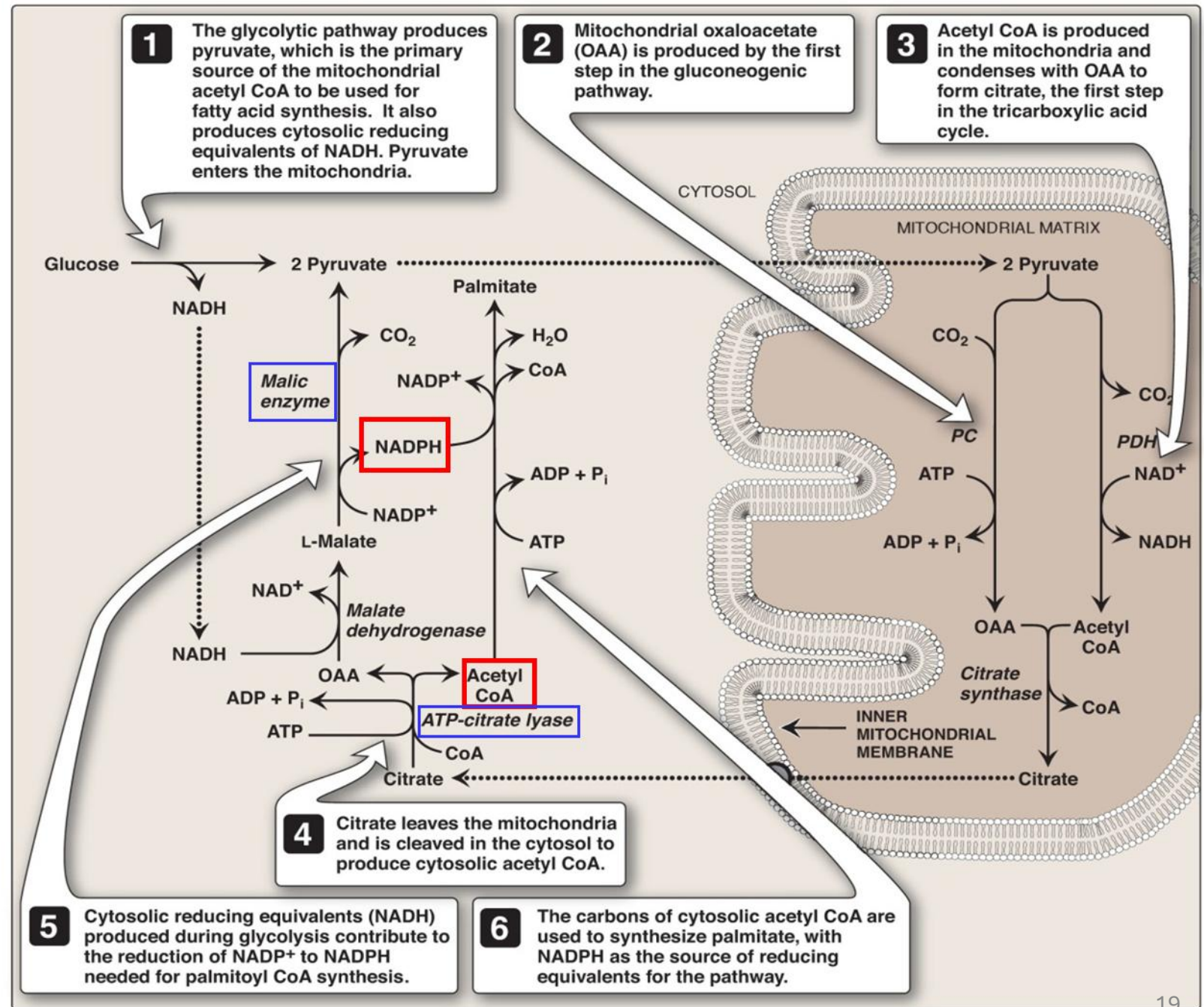
- Glucose 6-phosphate enters the pentose phosphate pathway (PPP) to generate NADPH in the cytosol.

2. Malic enzyme (NADP⁺ dependent malate dehydrogenase)

Reaction of
Malic Enzyme



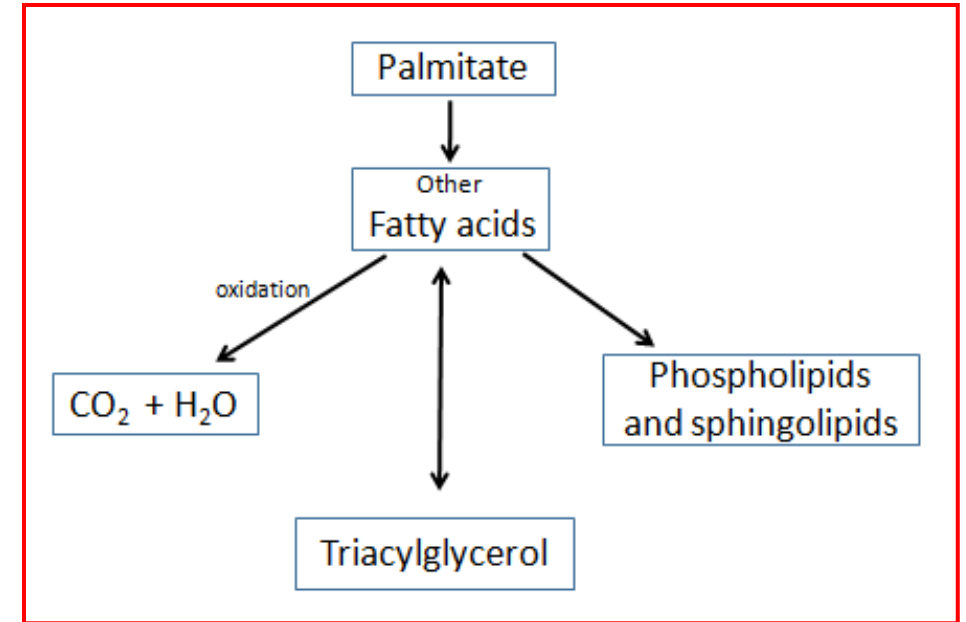
Transport of acetyl units and (NADPH) reducing equivalents used in FA synthesis



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Modification of Dietary and Endogenous Fatty Acids

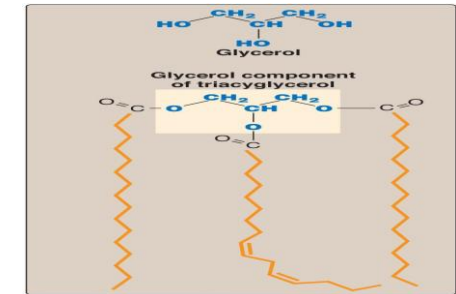
- **Palmitate** produced is **modified** to give rise to the other fatty acids.
- Fatty acids from diet may also be modified.



- **Modifications** include:
 - **Chain elongation** (endoplasmic reticulum) to produce longer fatty acids:
 - Elongation by enzymes known as **elongases** (different from FAS)
 - Brain makes **very-long-chain fatty acids** (VLCFAs 22C-24C) for synthesis of brain lipids.
 - **Desaturation**, producing unsaturated fatty acids: **Fatty acyl-CoA desaturases** introduce double bonds at positions **5, 6 or 9** in humans.

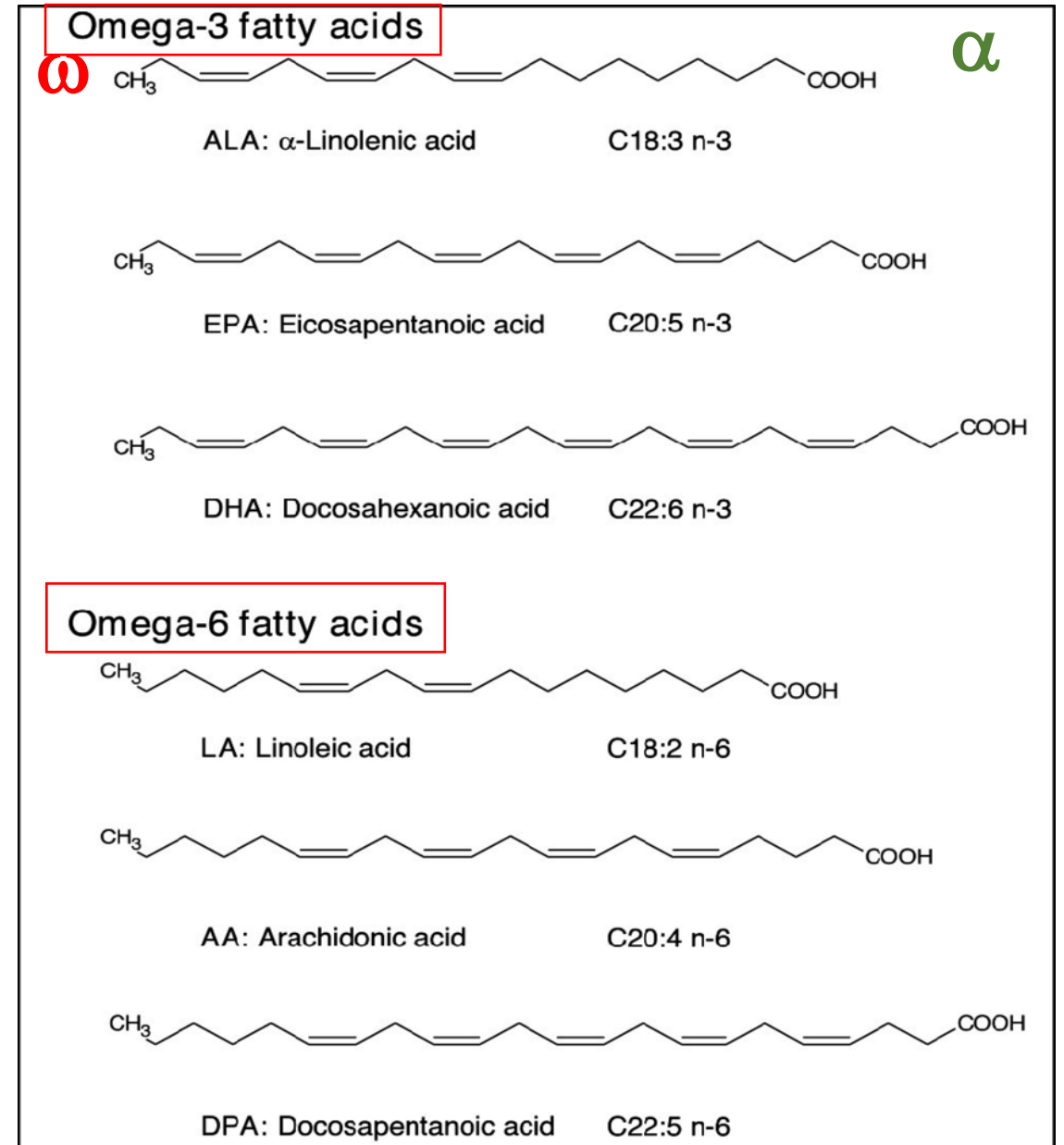
Essential Fatty Acids

- Mammals lack enzymes that can desaturate beyond the $\Delta 9$ position in the hydrocarbon chain
- Plants, including algae, can desaturate beyond the $\Delta 9$ position (ie. $\Delta 12$, $\Delta 15$ desaturases)
- Dietary essential fatty acids (cannot be synthesized in humans and must be provided in diet)
 - Linoleate (omega-6) and
 - α -linolenate (omega-3)



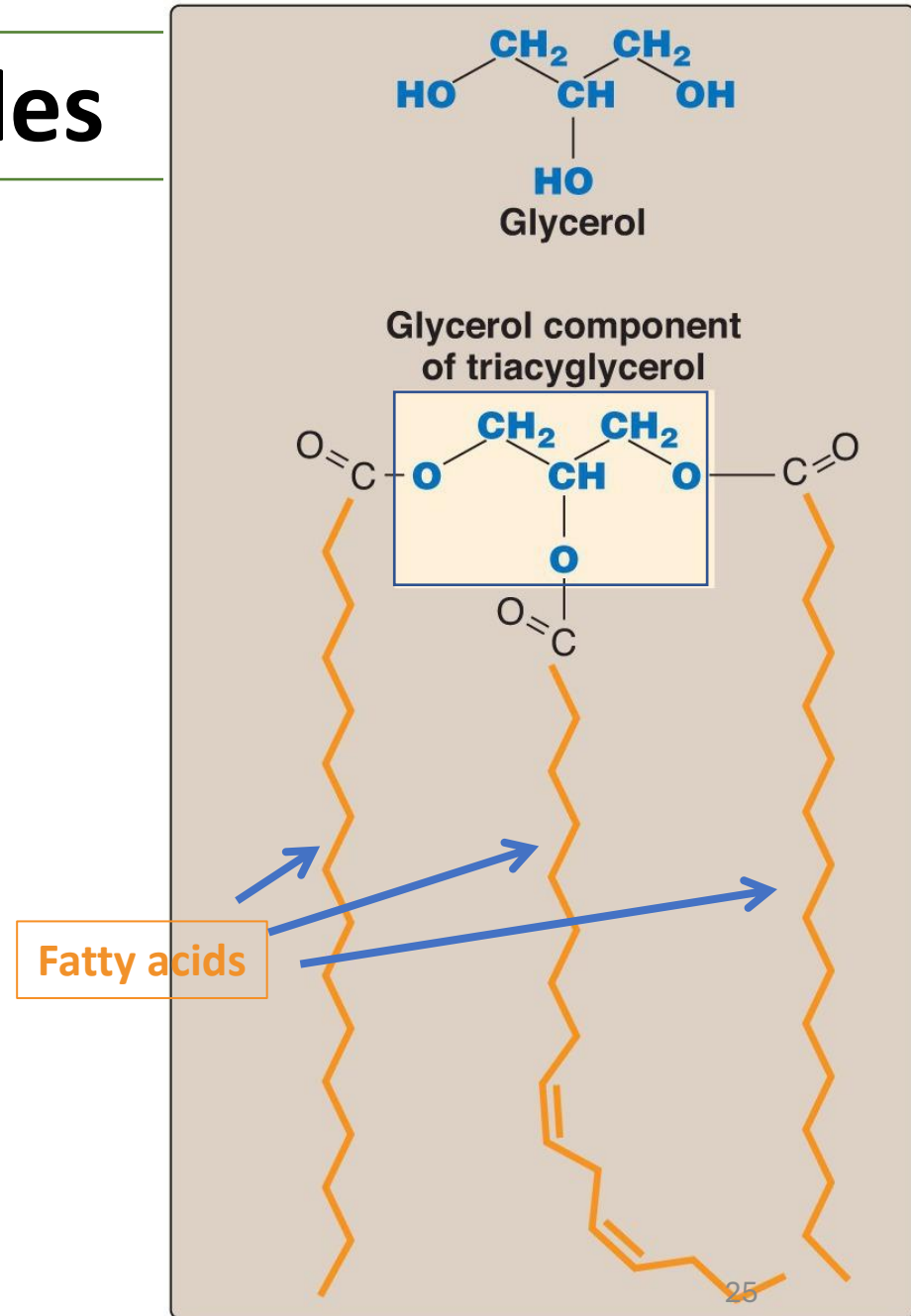
Essential Fatty Acids: Linolenic acid and Linoleic acid

- A fatty acid of $\omega 6$ family cannot be converted to a fatty acid of $\omega 3$ family.
- **Linolenic acid ($\omega 3$) $\rightarrow$ EPA $\rightarrow$ DHA**
 - **DHA** in **brain** phospholipids - provides high fluidity.
 - **DHA** is important for brain functions and the vision
- **Linoleic acid ($\omega 6$) $\rightarrow$ Arachidonic acid**
 - Arachidonic acid used to form eicosanoids (prostaglandins, thromboxane, leukotrienes).



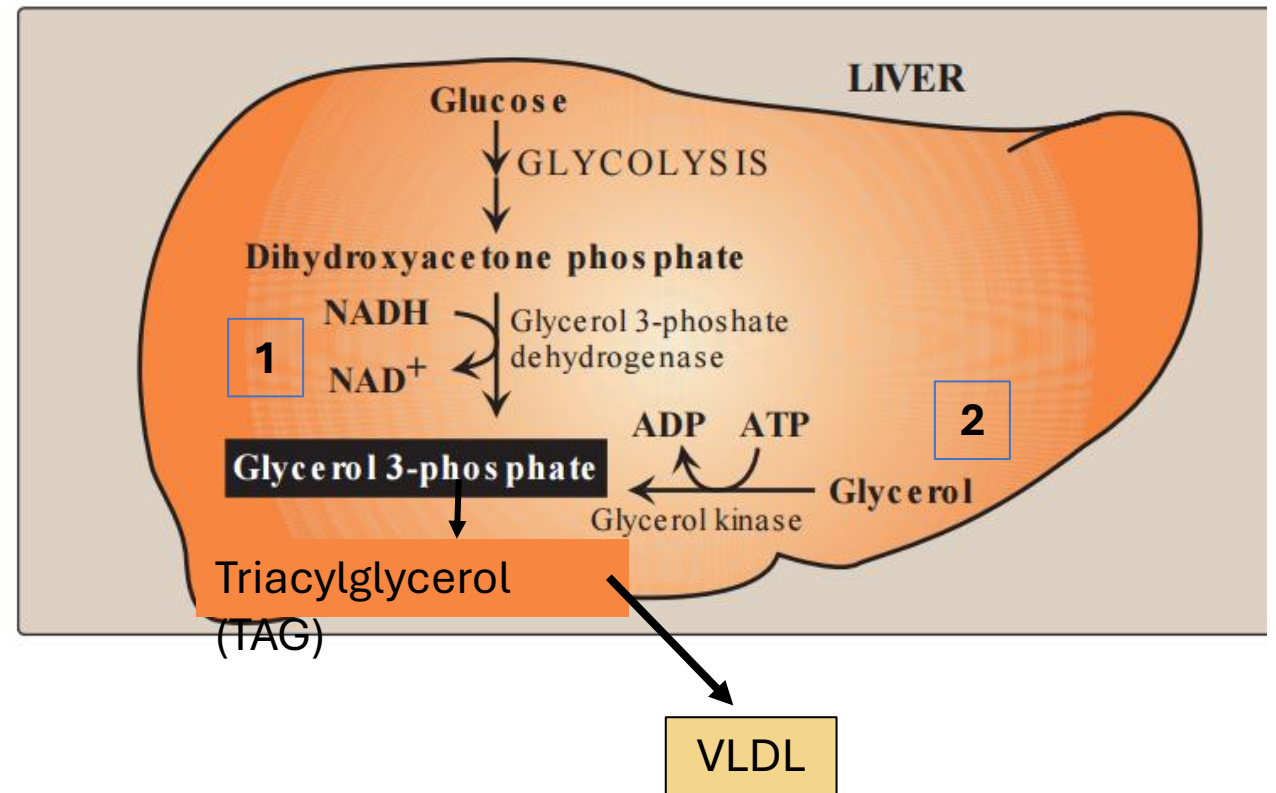
Triacylglycerol (TAG)/ Triglycerides

- **FA** are stored as **TAGs** (triglycerides)
- Glycerol + 3 fatty acids
- Glycerol 3-phosphate produced in liver and adipose tissue.
- **TAGs** synthesis: **Liver, adipose tissue** and in **intestinal mucosa**.
- Secretion of VLDL from the liver is an important mechanism to prevent the accumulation of TAG in the liver.

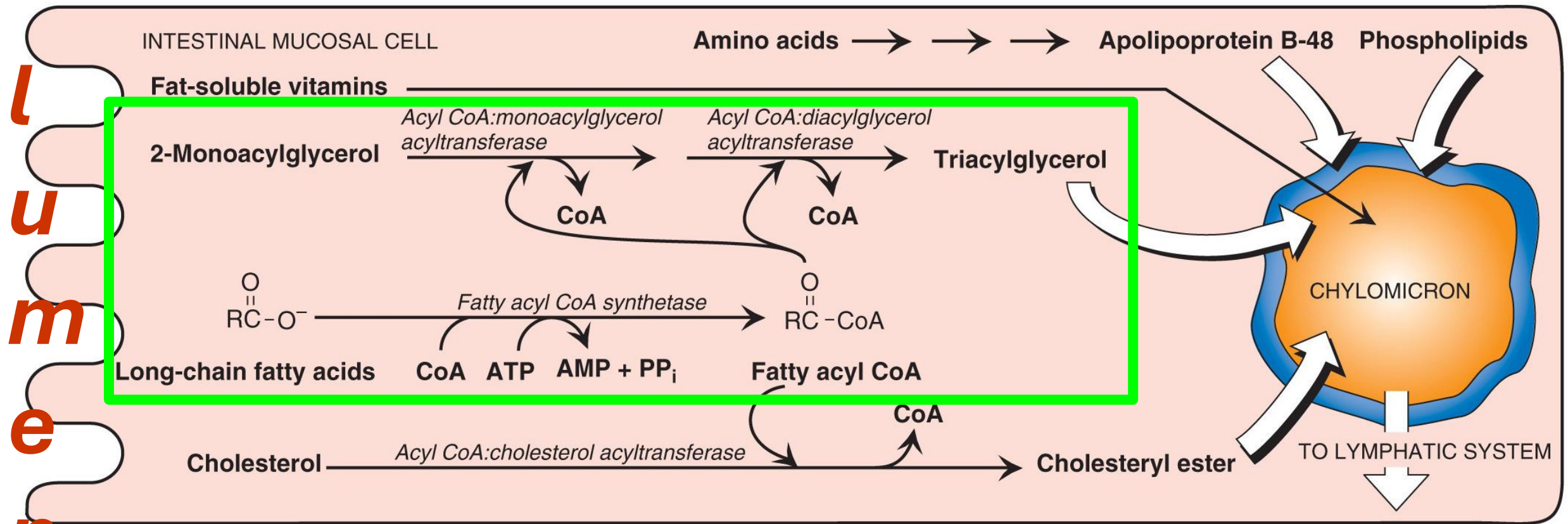


TAG Synthesis in the Liver

- Source of glycerol 3-phosphate:
 - **Dihydroxyacetone phosphate** (glycolysis) in liver and adipose tissue
 - **Glycerol** forms glycerol 3-phosphate by glycerol kinase (Liver)
 - Glycerol kinase present in liver, but absent in adipose tissue
- Three fatty acids added sequentially to form Triacylglycerol (TAG).
- TAG formed in the liver is packaged with apolipoproteins to form VLDL.
 - **Excessive storage of TAG in liver results in fatty liver disease**
- VLDL secreted into blood and transported to adipose and muscle where they deliver fatty acids to these tissues (Lipoprotein lipase).



Synthesis of TAGs in Intestinal mucosal cells (MAG Pathway)

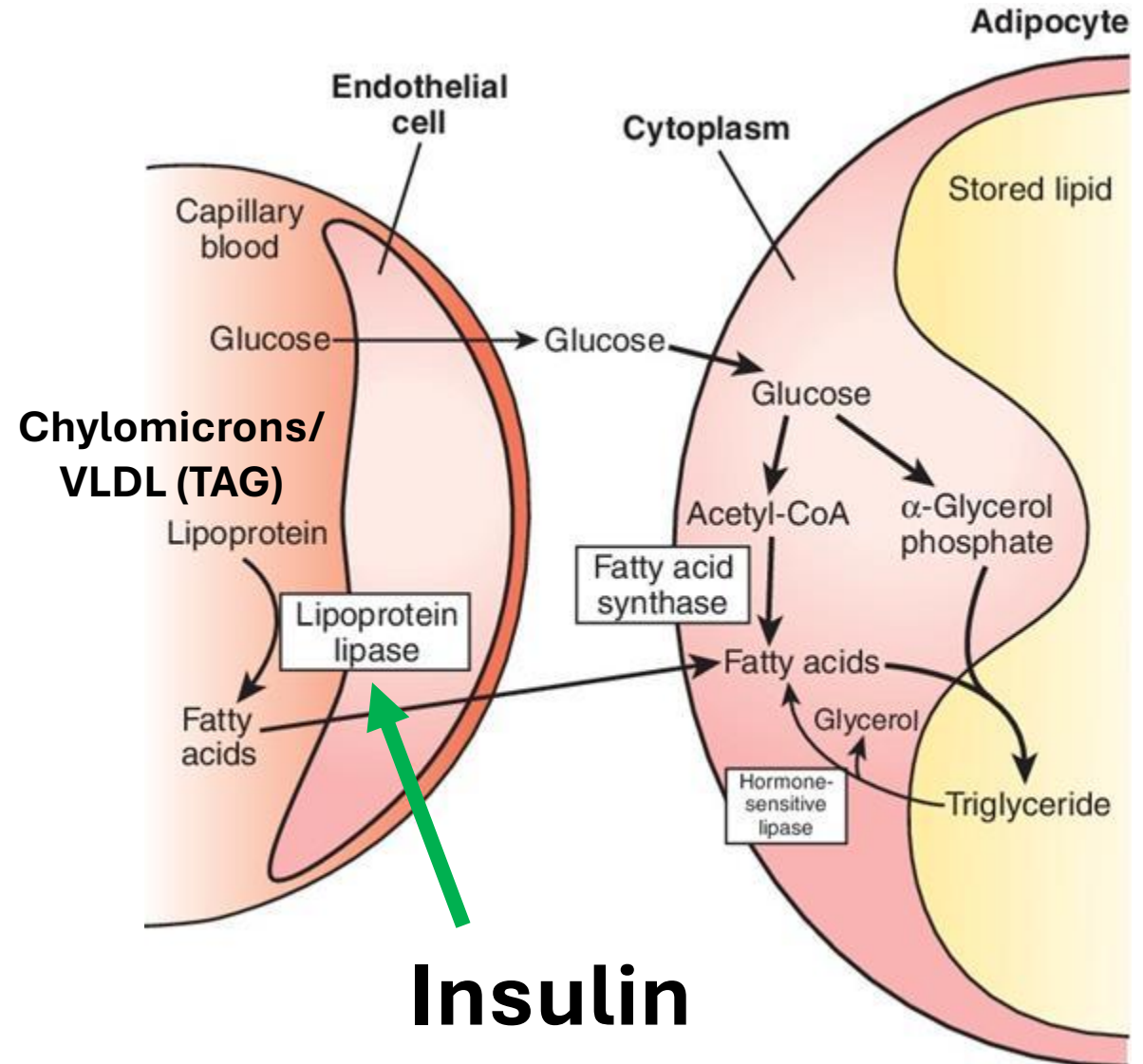


- **Monoacylglycerol (MAG):** End-product of lipid digestion
- Two Fatty acids added to form Triacylglycerol
- TAG released as **chylomicron** → Deliver fatty acids to adipose and muscle (lipoprotein lipase)

Lippincott Biochemistry: p. 195

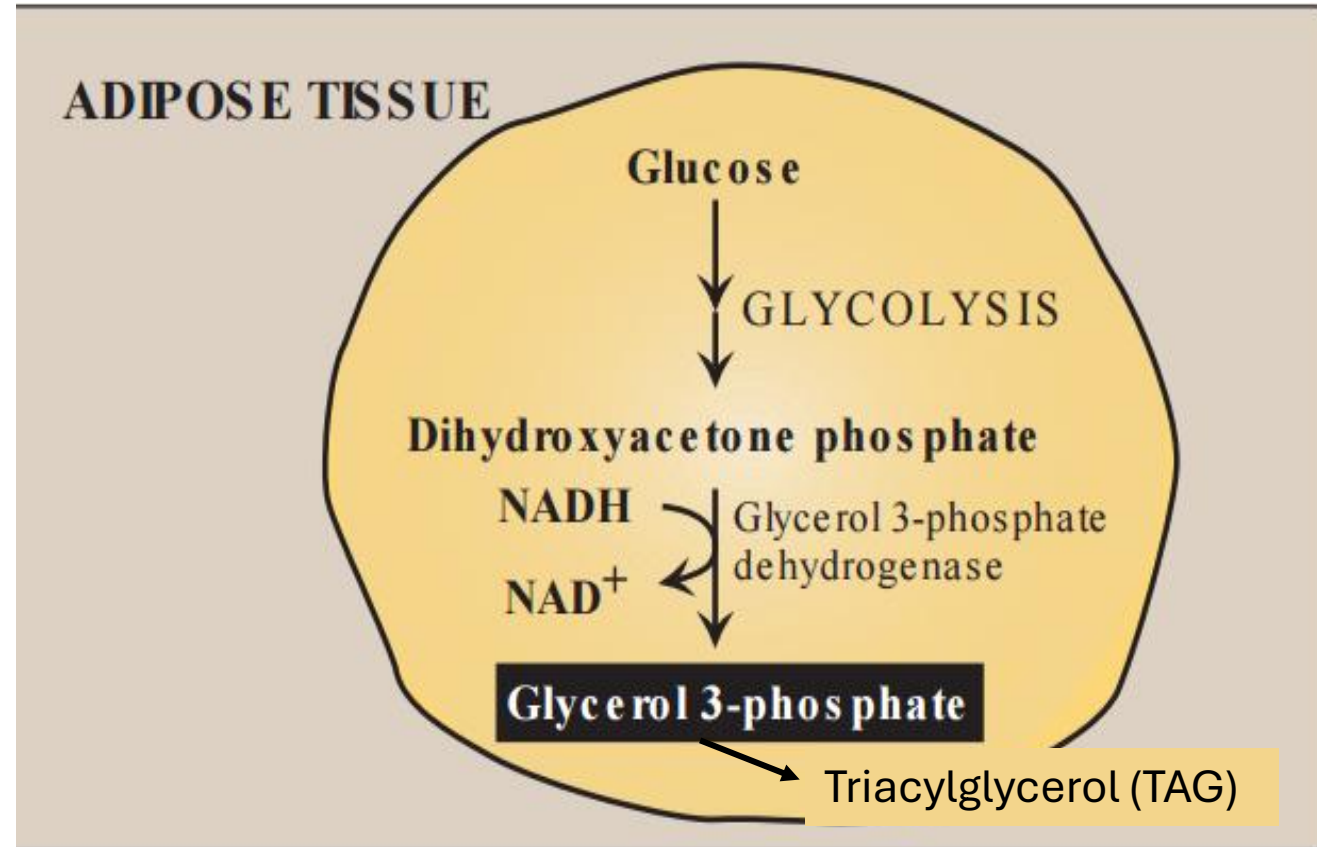
Delivery of fatty acids to adipocytes: Lipoprotein lipase

- Triglycerides in chylomicrons and VLDL hydrolyzed by **endothelial Lipoprotein lipase** → Fatty acids enter adipocytes → Stored as triglycerides (TAG)
- Lipoprotein lipase activated by **insulin**
- **Insulin facilitates triglyceride (TAG) storage**



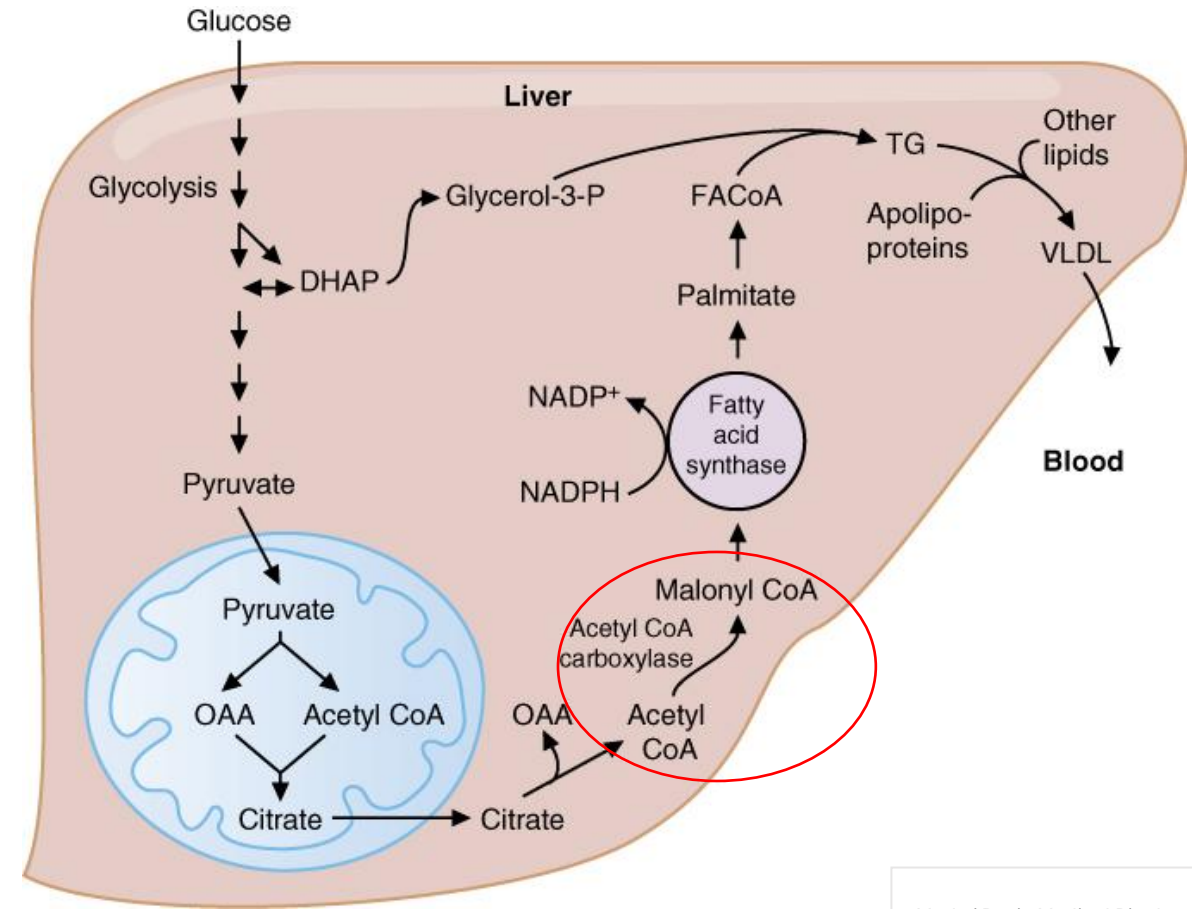
TAG synthesis in adipose tissue

- Source of glycerol 3-phosphate:
 - **Dihydroxyacetone phosphate** (glycolysis) in liver and adipose tissue
 - Glycerol kinase absent in adipose tissue
- Three fatty acids added sequentially to form Triacylglycerol (TAG).
- TAG stored in white adipocytes serve as the major energy reserve of the body.
 - **This energy store when in excess results in obesity**
- Fatty acids are released by **lipolysis (fasting/ exercise/stress)**
- TAG in brown adipocytes serves as source of heat through non-shivering thermogenesis.



Regulation of *de novo* Synthesis of Fatty Acids

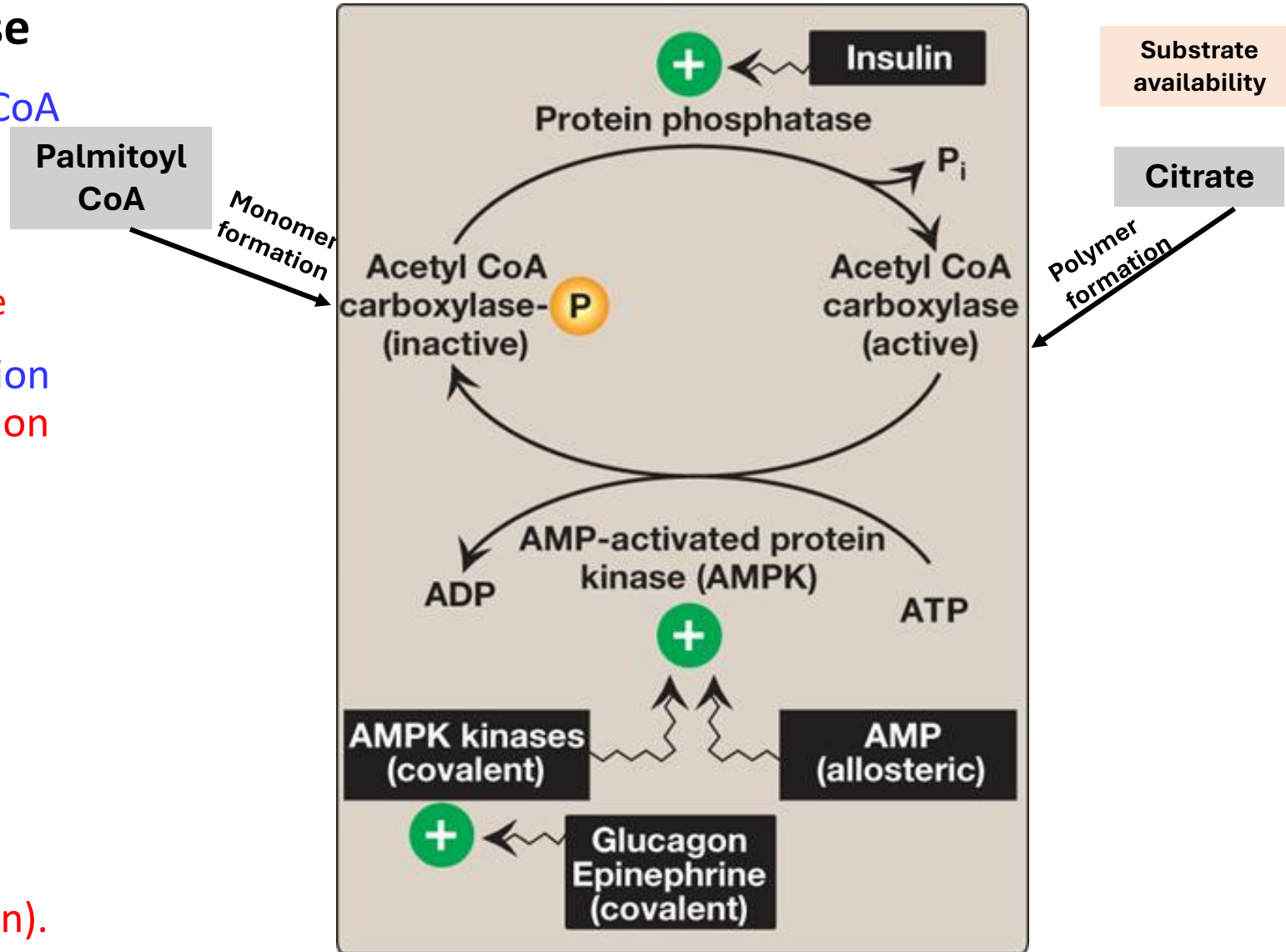
- **Acetyl CoA carboxylase (ACC)** is the rate-limiting enzyme in FA biosynthesis.
- The first committed step of fatty acid biosynthesis is catalyzed by **Acetyl CoA carboxylase (ACC)**, a biotin dependent enzyme.



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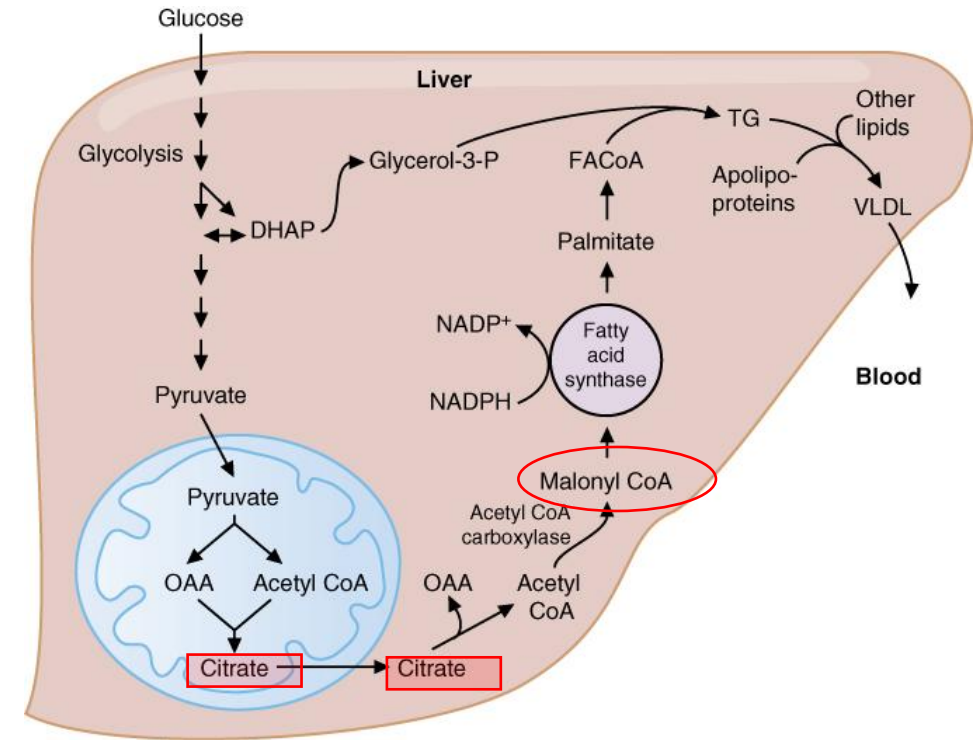
Regulation of *de novo* synthesis of Fatty acids

- Regulation of acetyl CoA carboxylase
 - **Substrate availability:** cytosolic acetyl CoA
 - **Allosteric:** activated by citrate and **inhibited by palmitoyl CoA and AMP.**
 - **Polymer** is active; **Monomer** is inactive
 - **Covalent:** activated by dephosphorylation (insulin) and **inhibited by phosphorylation (glucagon)**
 - **Enzyme induction** by insulin
- Is *de novo* synthesis of Fatty acids active in the fed state or during fasting?
 - Active in fed state **high blood glucose (insulin)** and
 - Inhibited at low blood glucose (glucagon).



Acetyl CoA Carboxylase (ACC): Feed-forward activation by citrate

- **Citrate** is found in **cytosol** when mitochondrial citrate levels are very high.
 - Cytosolic citrate is used for *de-novo* synthesis of fatty acid (and cholesterol synthesis).
- **Cytosolic citrate activates ACC** and leads to **polymer formation**.

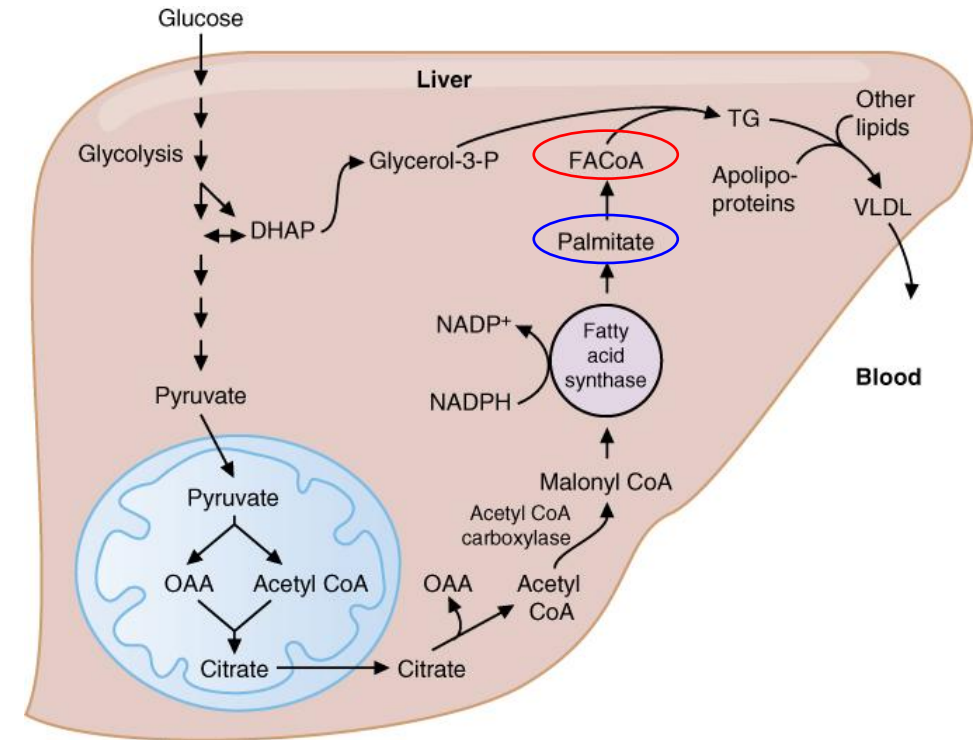


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Acetyl CoA Carboxylase (ACC):

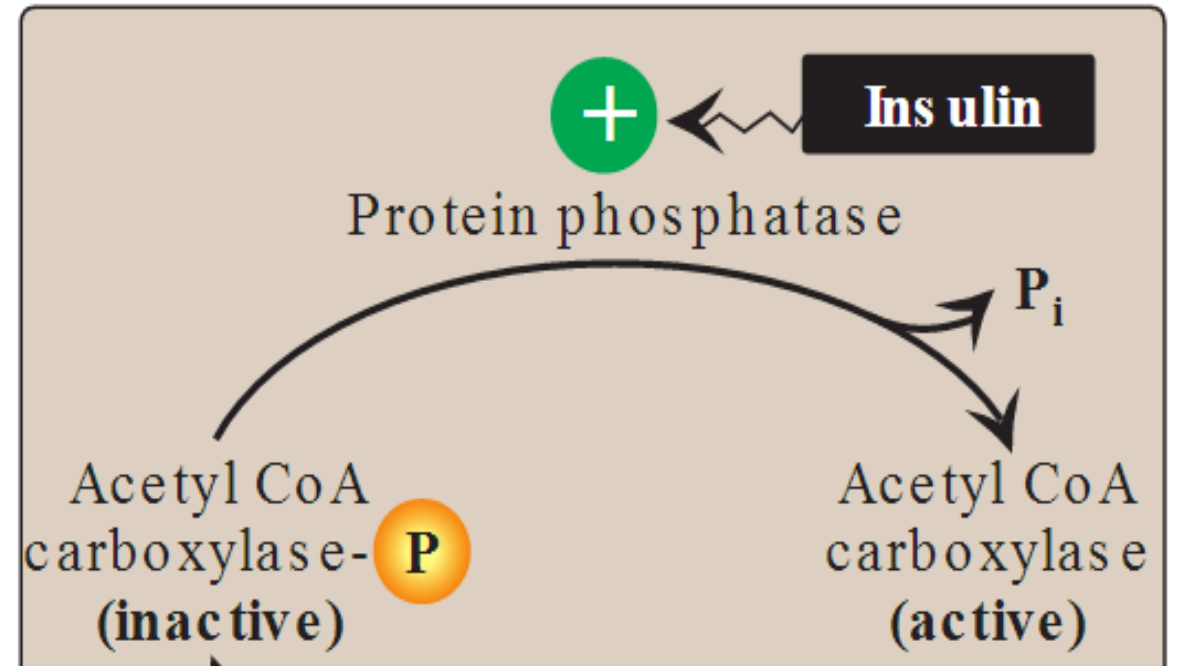
Feedback inhibition by fatty acyl CoA (palmitoyl CoA)

- Accumulation of long-chain fatty acyl CoA (end-product) (**palmitoyl CoA**) allosterically inhibits ACC
- Favors **monomer** formation (Less active)
- Hence, **fatty acid synthesis slows down**, so that less palmitate is formed.



Activation of acetyl CoA carboxylase by covalent modification

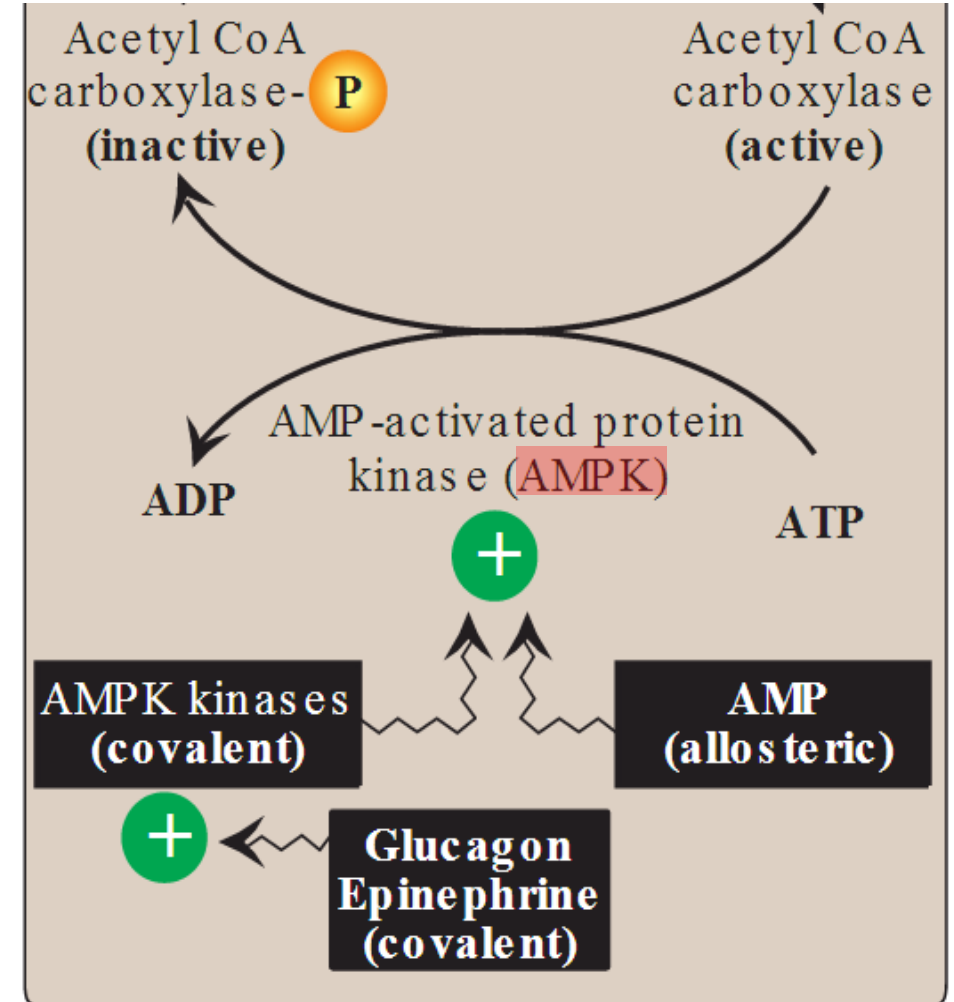
- Acetyl CoA carboxylase active when dephosphorylated.
- In the liver, Insulin favors *de-novo* synthesis of fatty acids.
- Insulin causes de-phosphorylation → Activates phosphoprotein phosphatase → Active acetyl CoA carboxylase
- Insulin also **induces** acetyl CoA carboxylase



Lippincott's p204

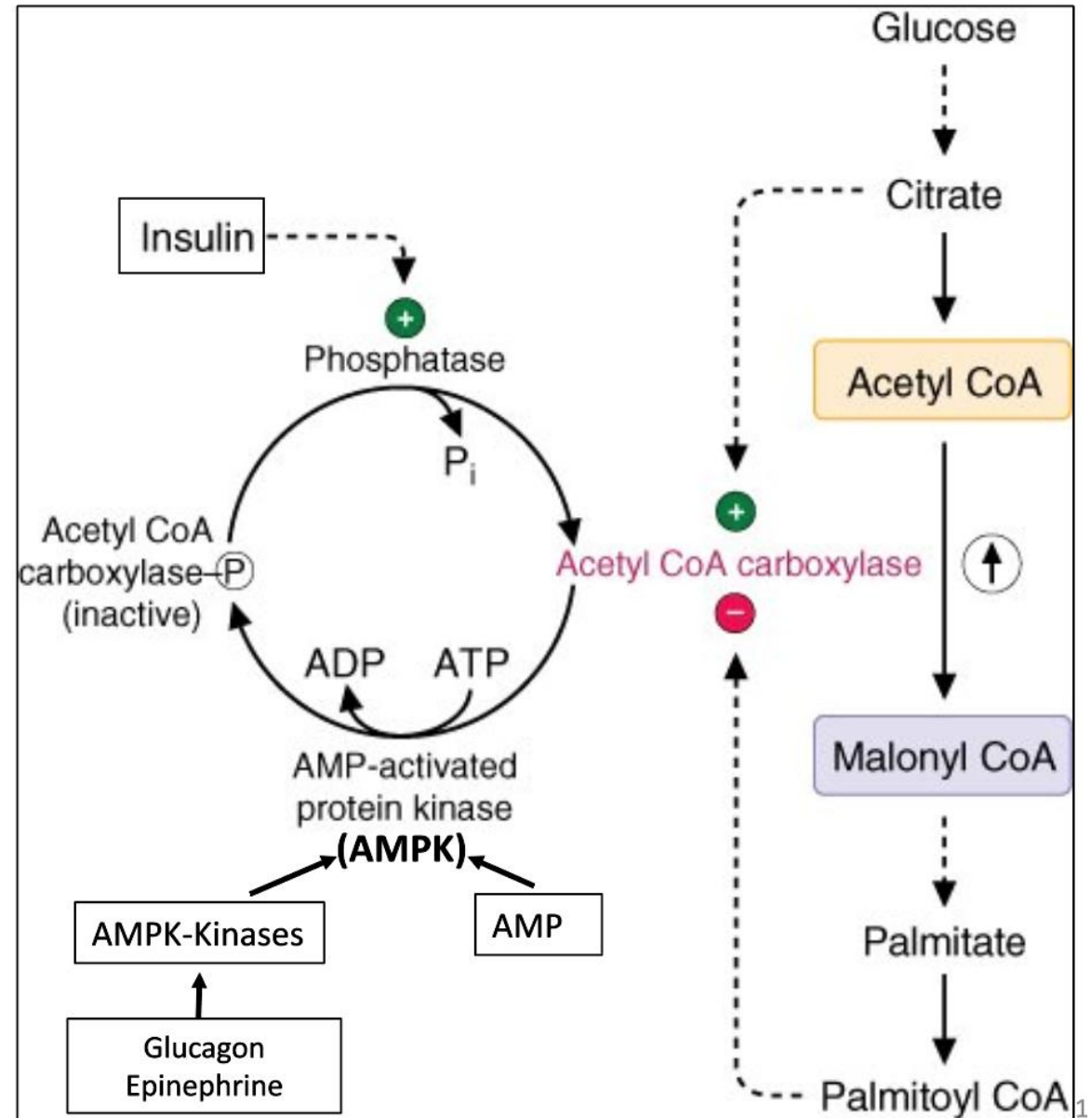
Inhibition of Acetyl CoA carboxylase

- **ACC inhibited** when **phosphorylated** by **AMP-activated protein kinase (AMPK)**
- **Activation of AMPK:**
 1. Allosteric by **AMP** (low energy)
 2. **Glucagon, epinephrine/norepinephrine** → Activation of AMPK → Phosphorylation of ACC (inactive)



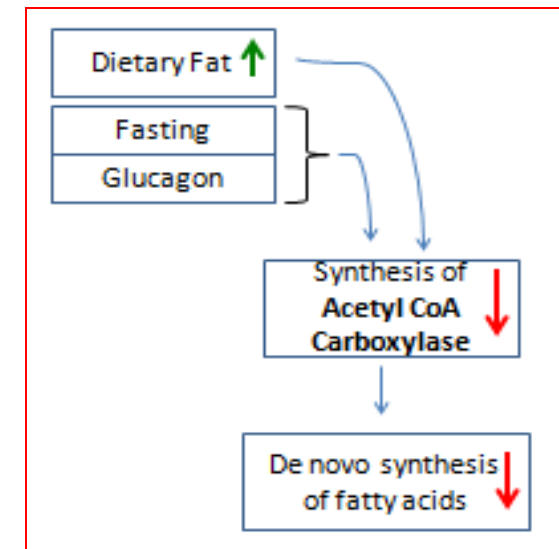
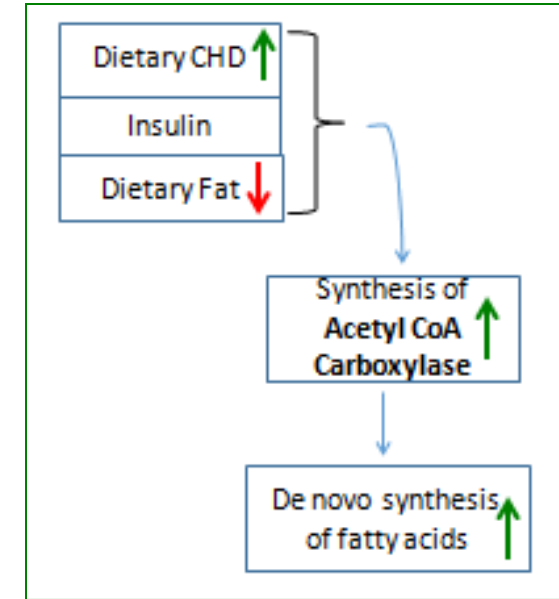
Short-term Regulation of Acetyl CoA Carboxylase - Summary

- **Substrate:** cytosolic citrate
- **Allosteric:** activated by citrate (polymer formation) and inhibited by palmitoyl CoA (monomer formation).
- **Covalent:** activated by dephosphorylation (insulin) and inhibited by phosphorylation (glucagon, epinephrine).
- **Enzyme induction** by insulin



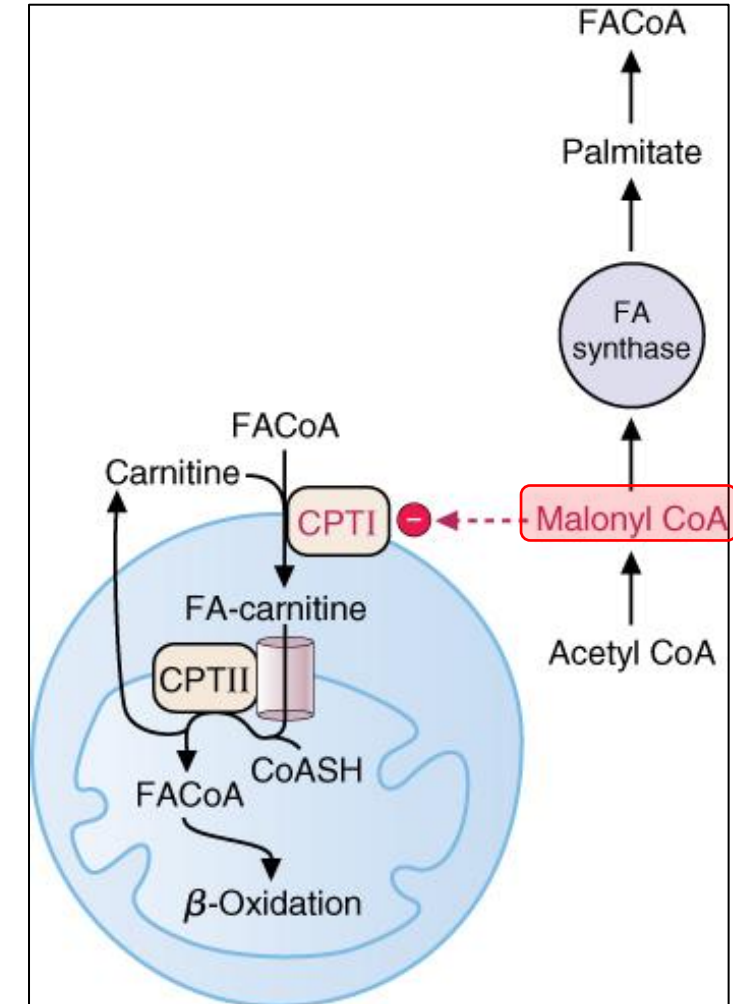
Long-term regulation of acetyl CoA carboxylase in *de novo* Synthesis of Fatty Acids

- **High carbohydrate and fat-free** diets and insulin lead to increased synthesis of ACC (**up-regulation, induction**) resulting in increased synthesis of FA
- **High-fat diets, fasting and glucagon** lead to decreased synthesis (**down-regulation, repression**) resulting in decreased synthesis of FA



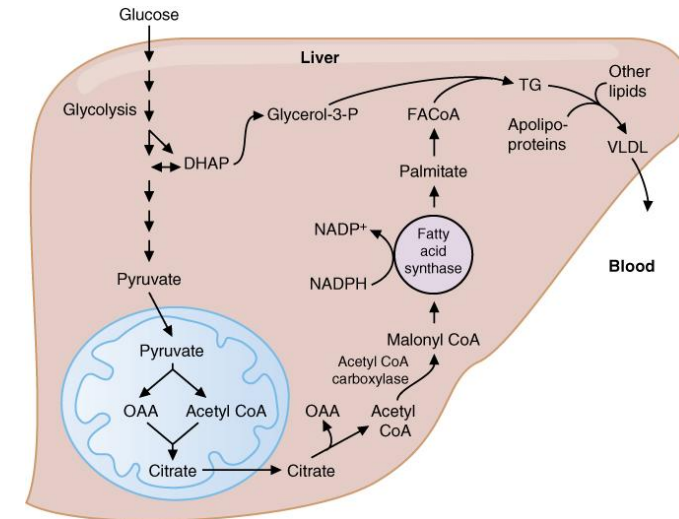
Malonyl-CoA Inhibition of Carnitine-Palmitoyl Transferase I (CPT I)

- **Malonyl-CoA** formed in **cytosol** during FA synthesis **inhibits CPT-I** which is involved in the transport of long-chain fatty acids into the **mitochondria** for β -oxidation.
 - Ensures that **during FA synthesis, β -oxidation inhibited (reciprocal regulation)**
 - Newly synthesized FA remain in the cytosol and are not immediately oxidized in the fed state



Summary

- Fatty acids are synthesized in **liver**, primarily from glucose.
- Glucose converted to pyruvate (glycolysis) → enters mitochondria and forms acetyl-CoA; Acetyl CoA and OAA → Citrate (Citrate synthase)
- Citrate transported to cytosol; Forms acetyl-CoA (by ATP-citrate lyase) – source of carbons for fatty acid biosynthesis
- Enzymes: **Acetyl-CoA carboxylase** (key regulatory step) and fatty acid synthase, produce palmitic acid (16 carbon, saturated)
- Insulin (well-fed state) activates fatty acid synthesis
- Palmitoyl-CoA can be elongated or desaturated (adding double bonds) by enzymes in endoplasmic reticulum.
- Fatty acids used to produce triacylglycerols (for energy storage in adipocytes) and glycerophospholipids and sphingolipids (for structural components of cell membranes)



Marks' Basic Medical Biochemistry

Fatty acid Synthesis - Summary

SYNTHESIS	
Greatest flux through pathway	After carbohydrate-rich meal
Hormonal state favoring pathway	High insulin/glucagon ratio
Major tissue site	Primarily liver
Subcellular location	Primarily cytosol
Carriers of acyl/acetyl groups between mitochondria and cytosol	Citrate (mitochondria to cytosol)
Phosphopantetheine-containing active carriers	Acyl carrier protein domain, coenzyme A
Oxidation/reduction coenzymers	NADPH (reduction)
Two-carbon donor/product	Malonyl CoA: donor of one acetyl group
Activator	Citrate
Inhibitor	Long-chain fatty acyl CoA (inhibits <i>acetyl CoA carboxylase</i>)
Product of pathway	Palmitate
Repetitive four-step process	Condensation, reduction, dehydration, reduction

Thank you!

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